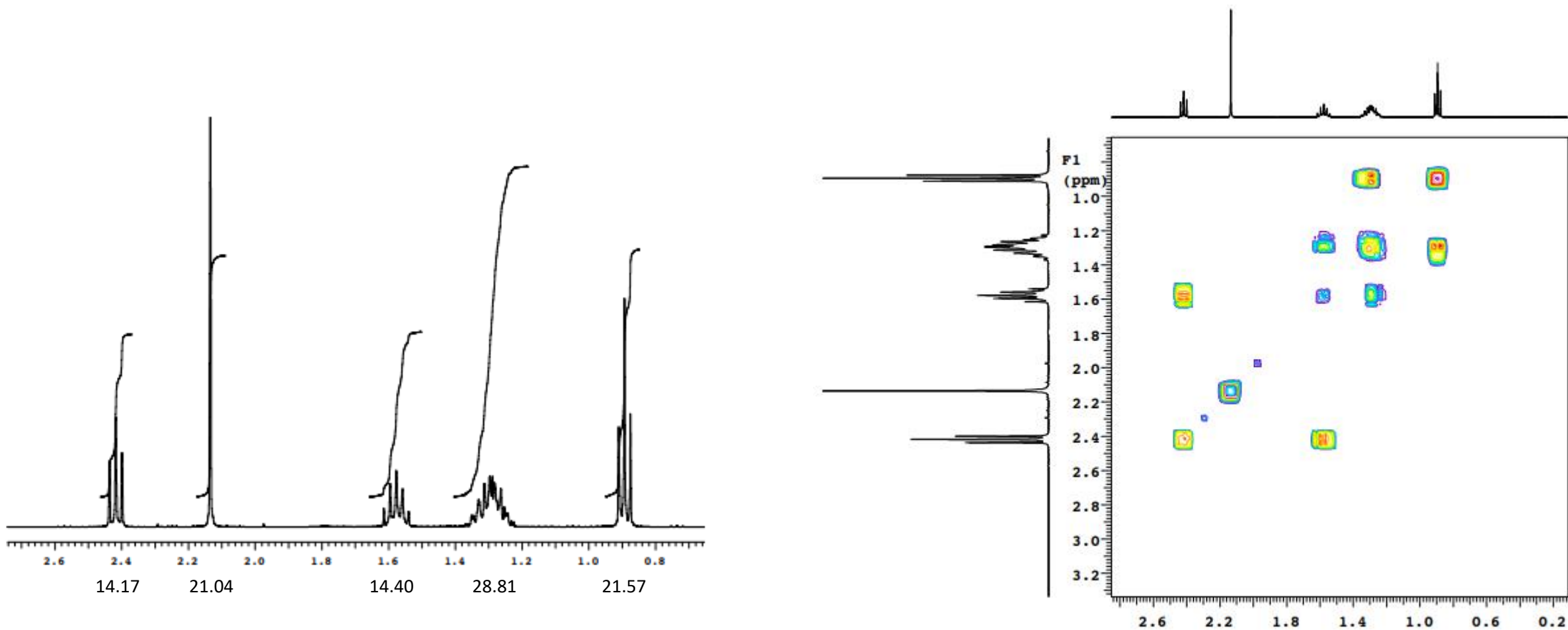


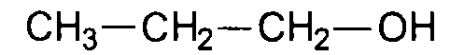
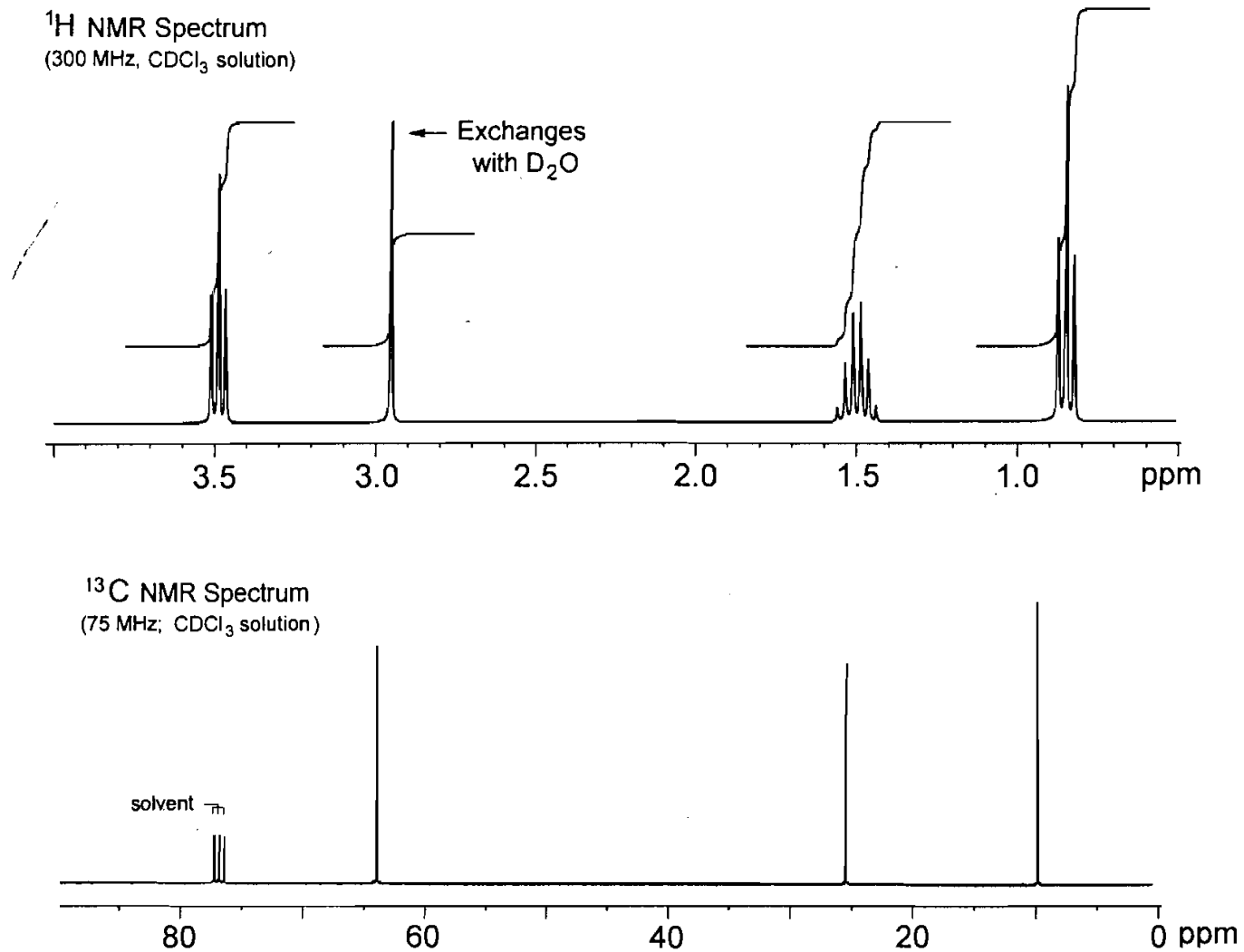
1. Predict the structure of the molecule based on the data given, the data doesn't indicate number of H, it indicates peak area of the protons. Explain the 2D NMR. The molecular formula of the compound is $C_7H_{14}O$.

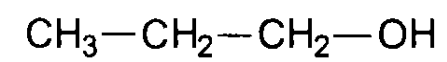
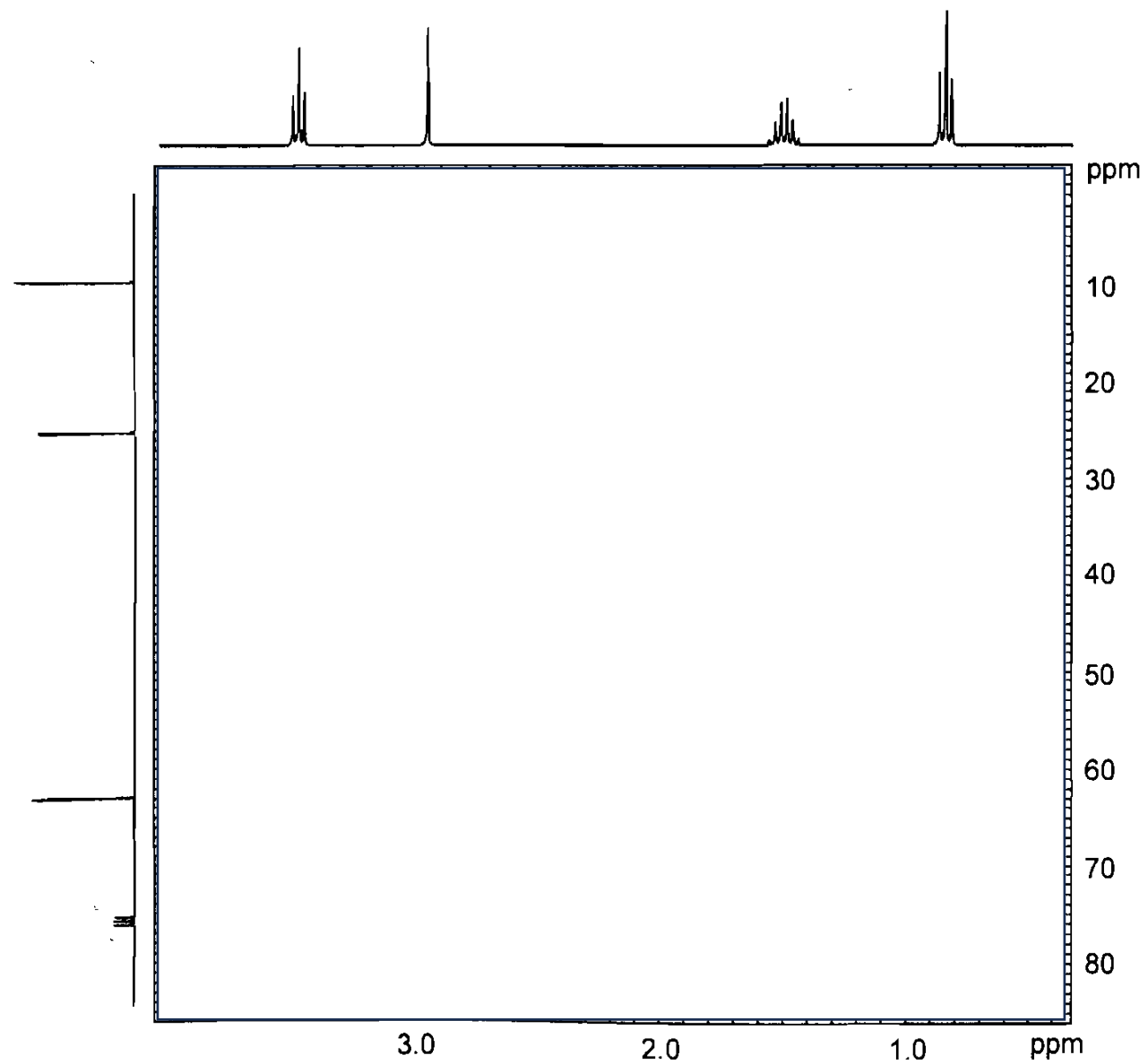
^{13}C : 207, 43, 31, 29, 22.8, 22.4, 14.1



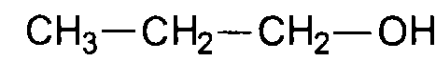
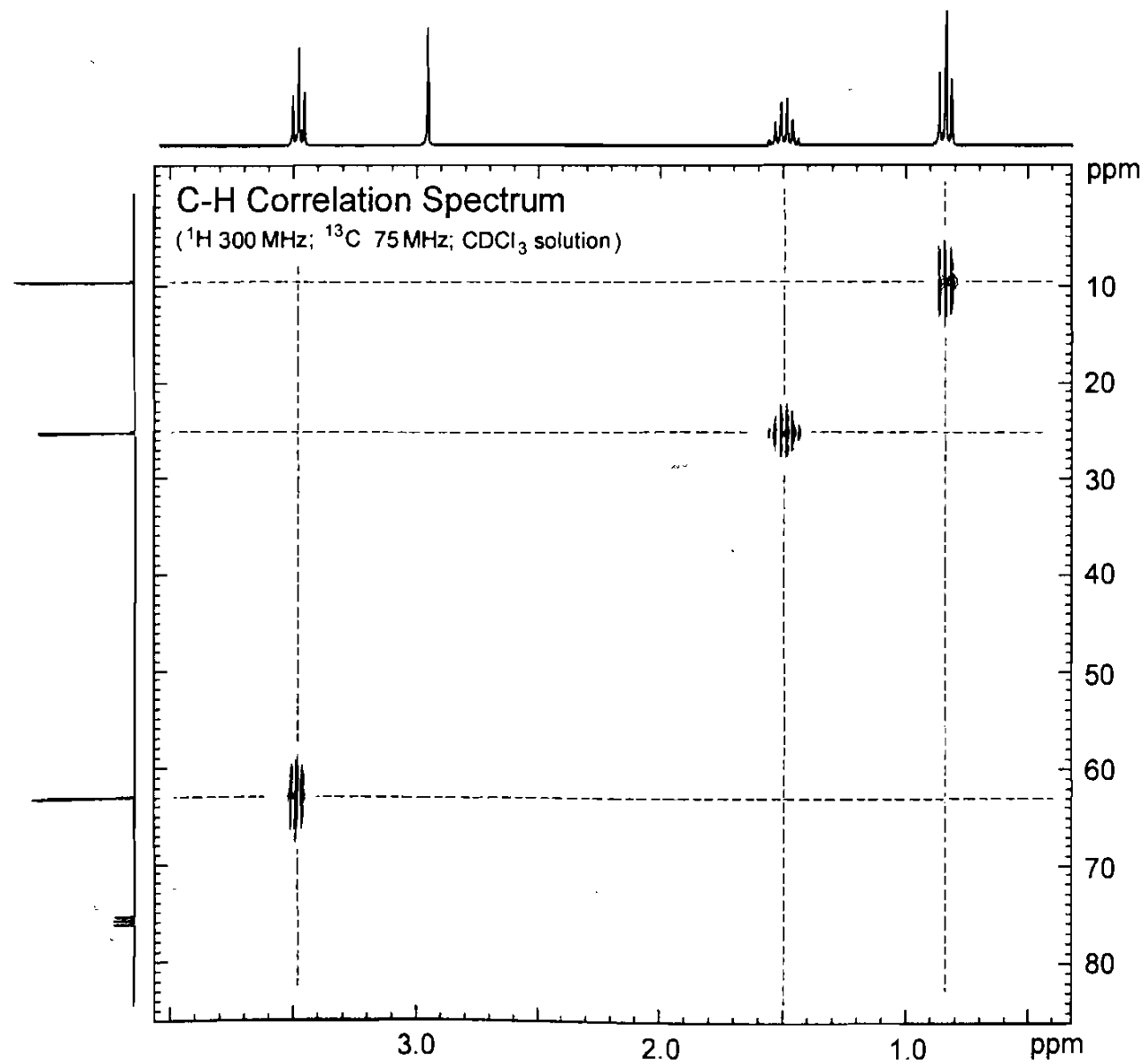
HETeronuclear **COR**relation (**HETCOR**)

^1H - ^{13}C correlations: ^1H NMR spectra on one axis and the ^{13}C NMR spectra on the other





No Diagonal peak



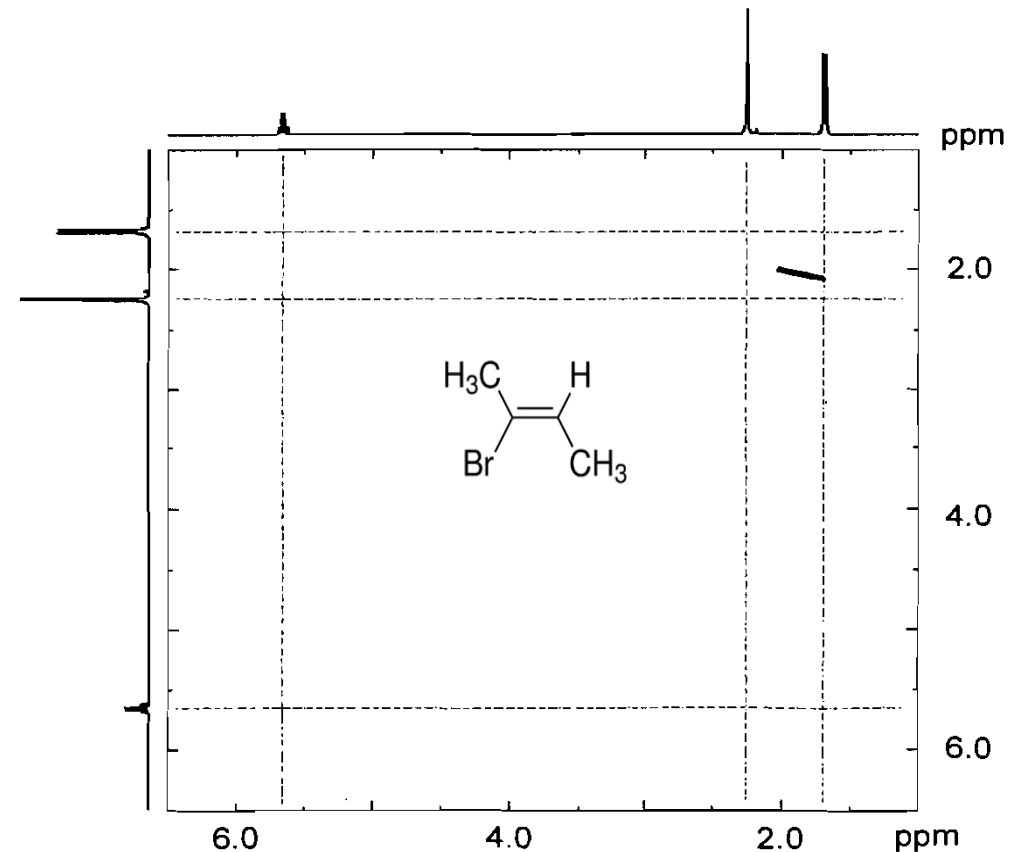
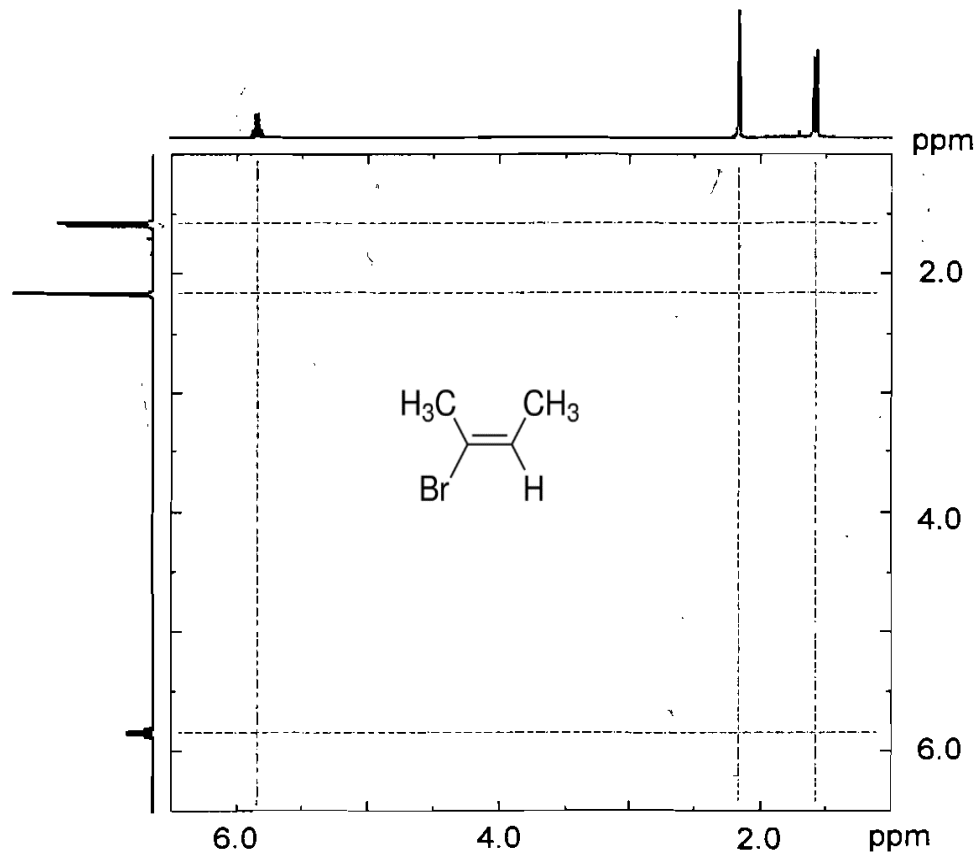
No Diagonal peak

Nuclear Overhauser Effect Spectroscopy, or **NOESY**, shows through-space interactions within the molecule, rather than the through-bond interactions seen in the other methods. This method is especially useful for determining stereochemical relationships in a molecule. In two stereoisomers, the atoms are all connected in exactly the same order, through exactly the same bonds.

Nuclear Overhauser Enhancement Spectroscopy (NOESY)

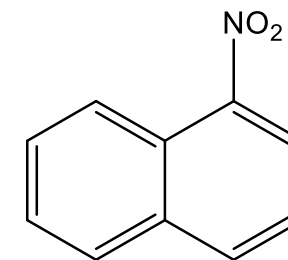
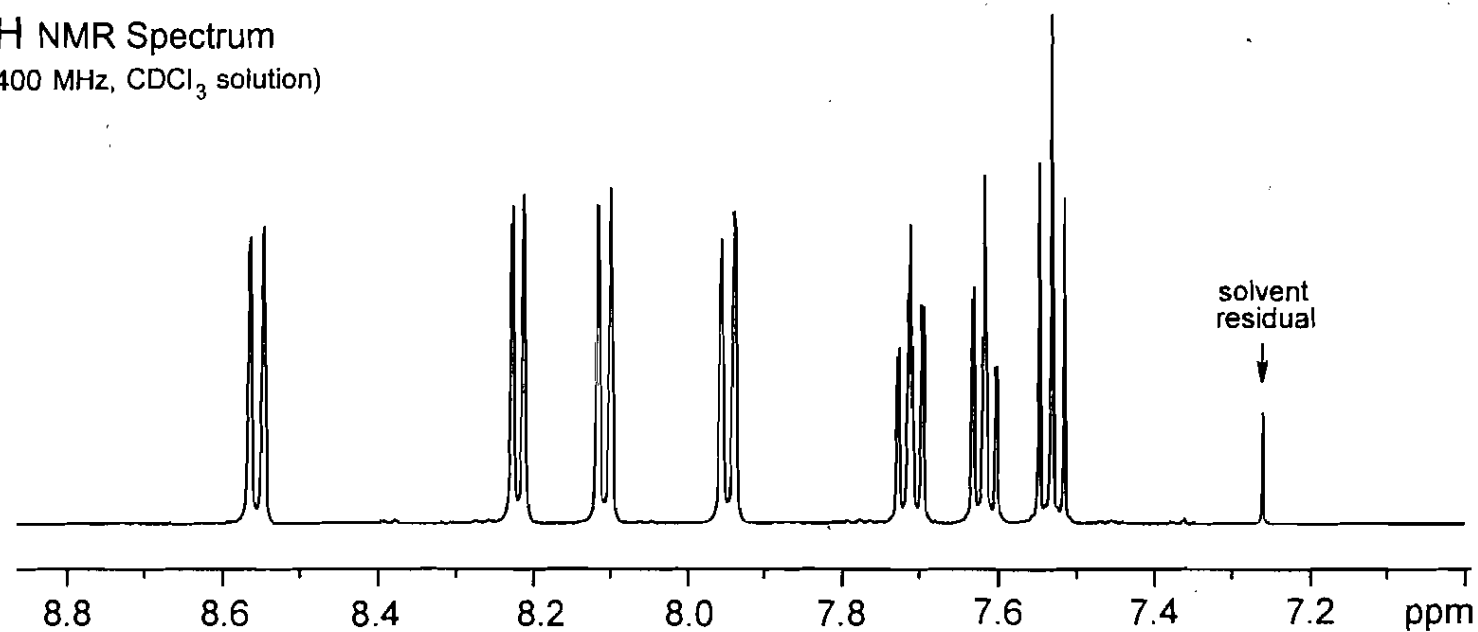
✓ ^1H - ^1H correlation through space (NOE works typically within 4-5 Å separation)

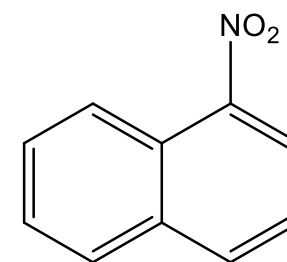
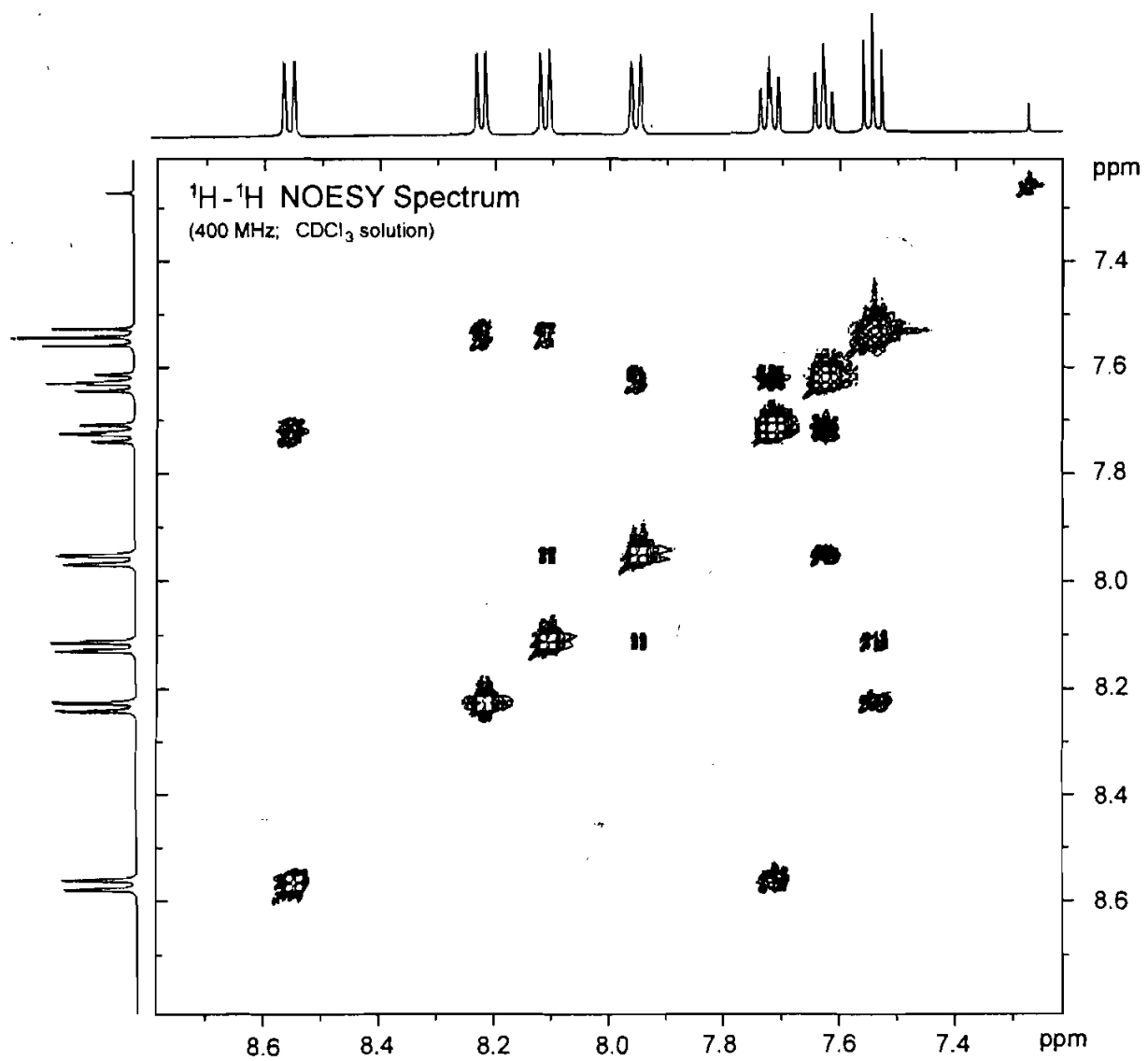
Example: The (*E*)- and (*Z*)-isomers of 2-bromo-2-butene ($\text{C}_4\text{H}_7\text{Br}$) are difficult to distinguish by ^1H NMR spectroscopy. Both isomers have 3 resonances one between 5.5 and 6.0 ppm, one between 2.0 and 2.5, and one between 1.5 and 2.0 ppm,-

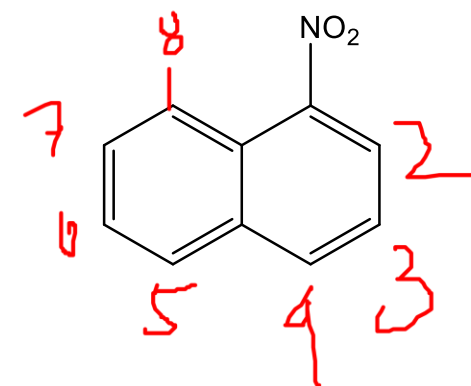
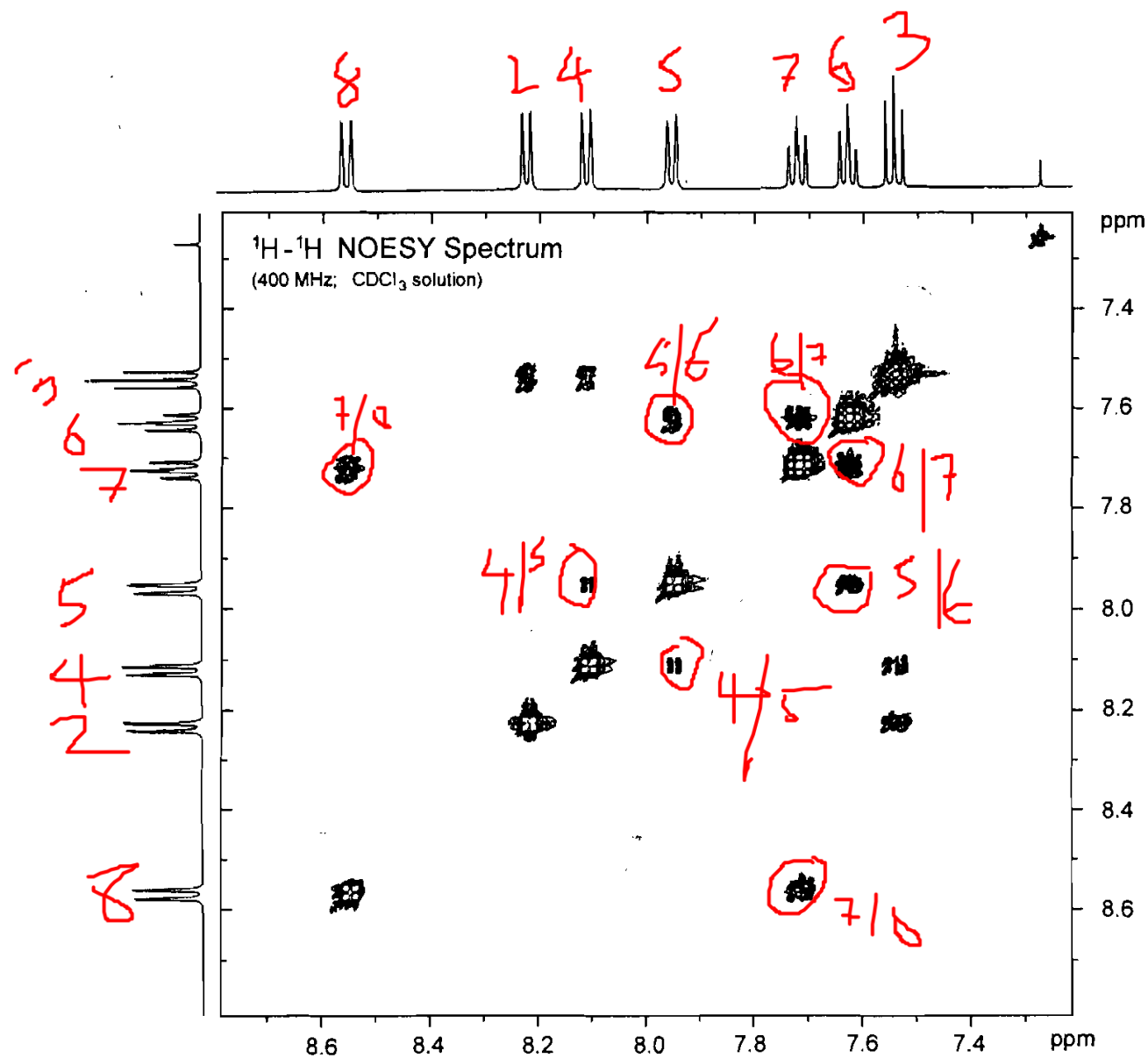


The ^1H NMR spectrum of 1-nitronaphthalene (recorded at 298K in CDCl_3) solution is given below. The 2-dimensional ^1H NOESY spectrum is also provided. Given that the nitro group at position 1 will always extensively deshield the proton at position 8 such that it will appear most downfield, use the NOESY spectrum to assign all of the other protons in the spectrum and show in a tabular form.

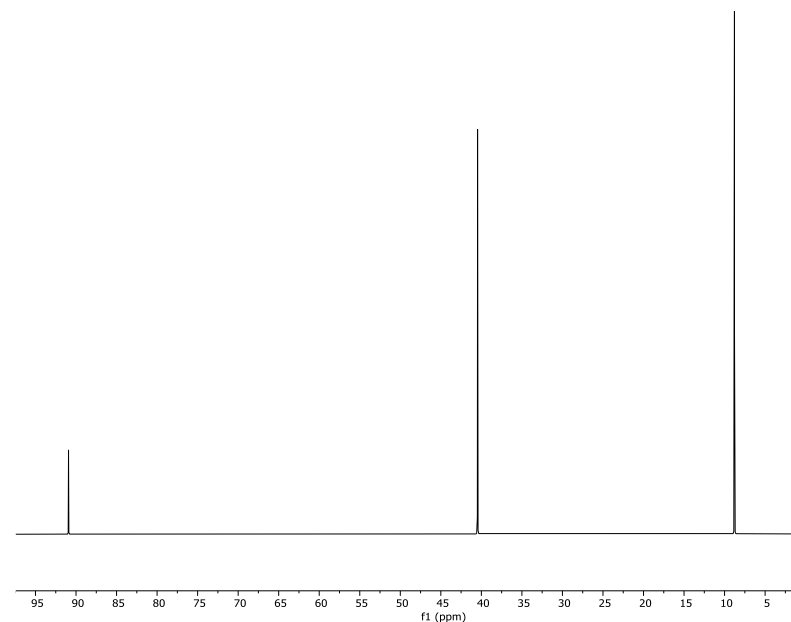
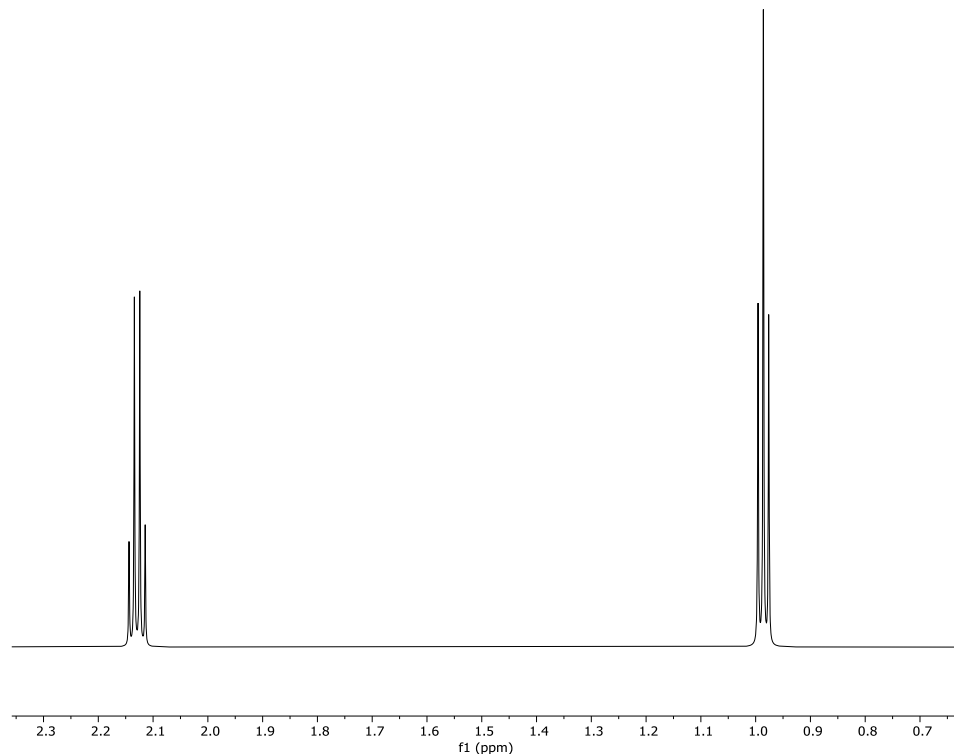
^1H NMR Spectrum
(400 MHz, CDCl_3 solution)







4. The ^1H and ^{13}C NMR data is provided of one compound having formula $(\text{C}_5\text{H}_{10}\text{Cl}_2)$. Predict the structure of the compound.



5. Two isomers with the formula $\text{C}_{10}\text{H}_{12}\text{O}$. Predict the structure of the compounds.

Compound A: IR (neat): 1675 cm^{-1} ; ^1H NMR (CDCl_3): 1.20 (d, $J = 7\text{ Hz}$, 6H), 3.53 (septet, $J = 7\text{ Hz}$, 1H), 7.94 (m, 2H), 7.68 (m, 1H), 7.3 (m, 2H).

Compound B: IR (neat): 1705 cm^{-1} ; ^1H NMR (CDCl_3): 0.95 (t, $J = 7\text{ Hz}$, 3H), 2.35 (q, $J = 7\text{ Hz}$, 2H), 3.6 (s, 2H), 7.20 (m, 5H).

Q 6. Match the following with ^{13}C NMR value:

Compound

^{13}C NMR chemical shift (δ ppm)

(A) Acetic acid

(i) 95

(B) Acetonitrile

(ii) 115

(C) Acetone

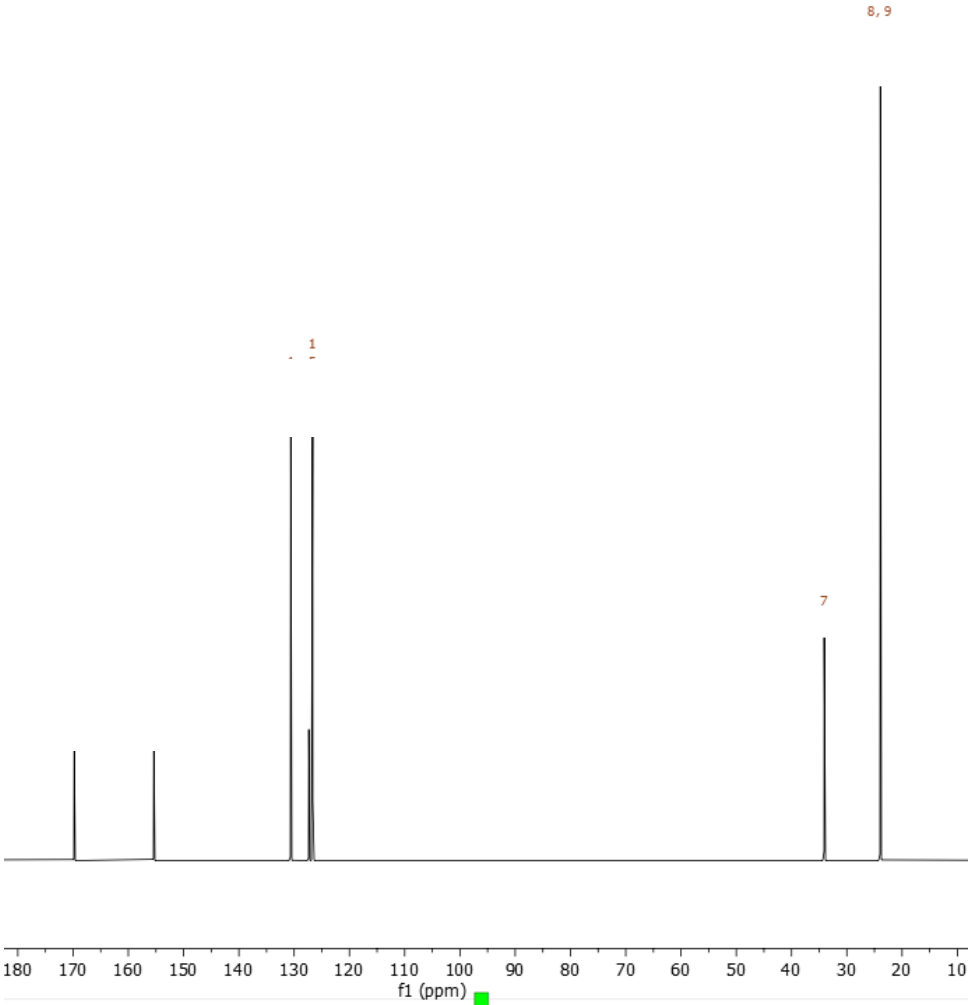
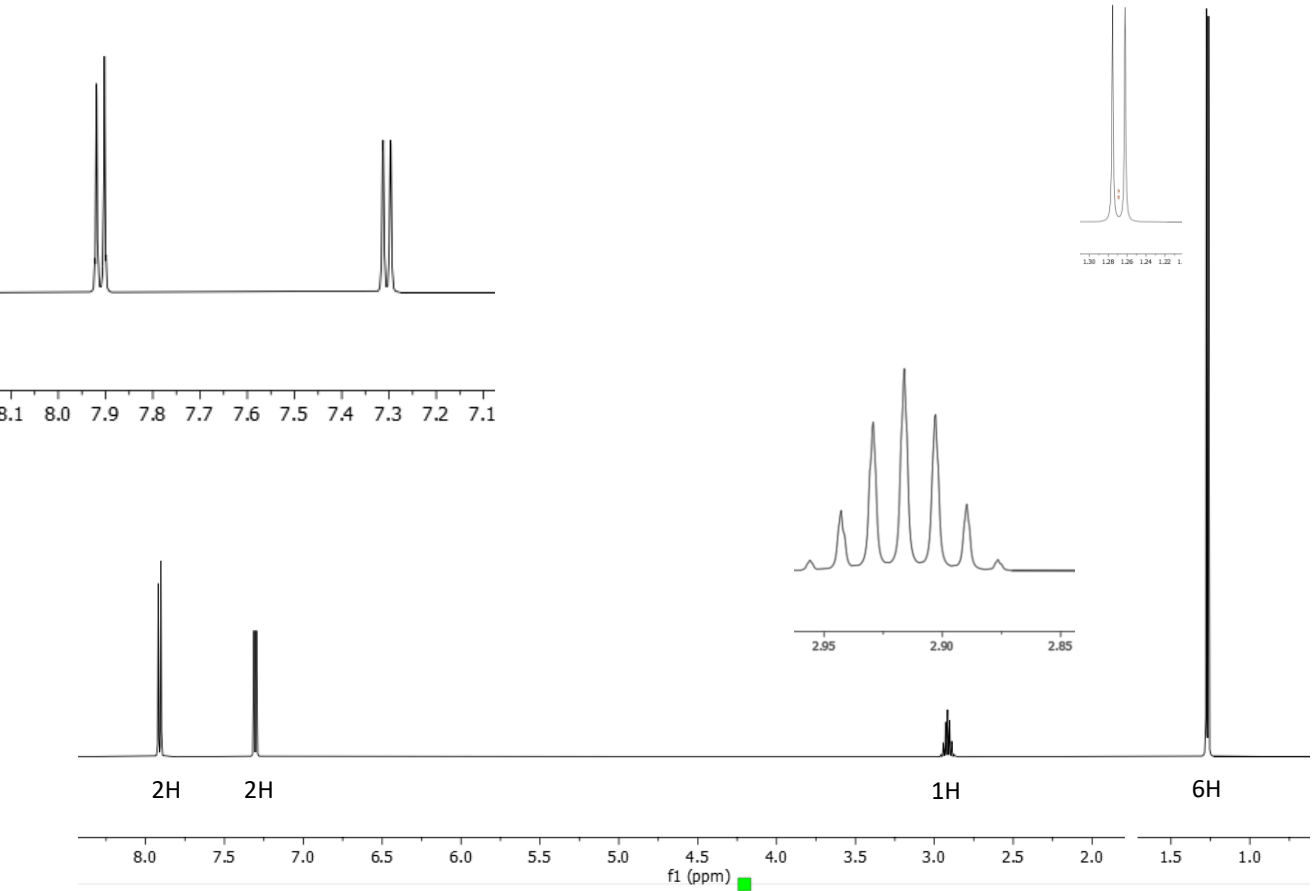
(iii) 175

(D) Carbon tetrachloride

(iv) 205

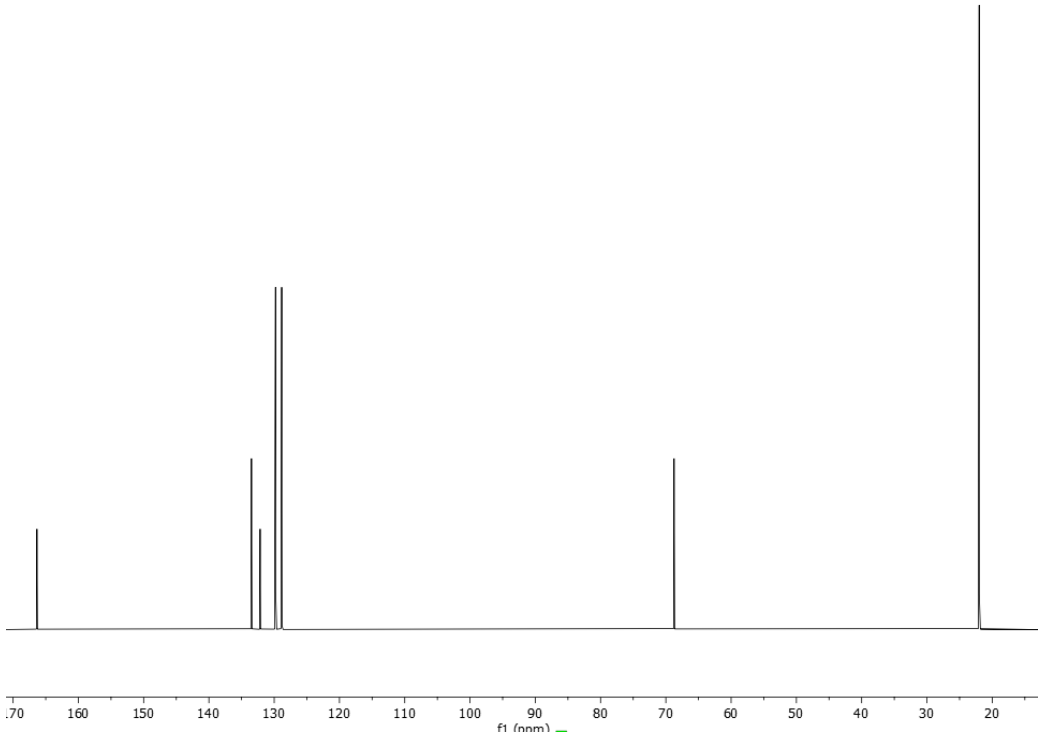
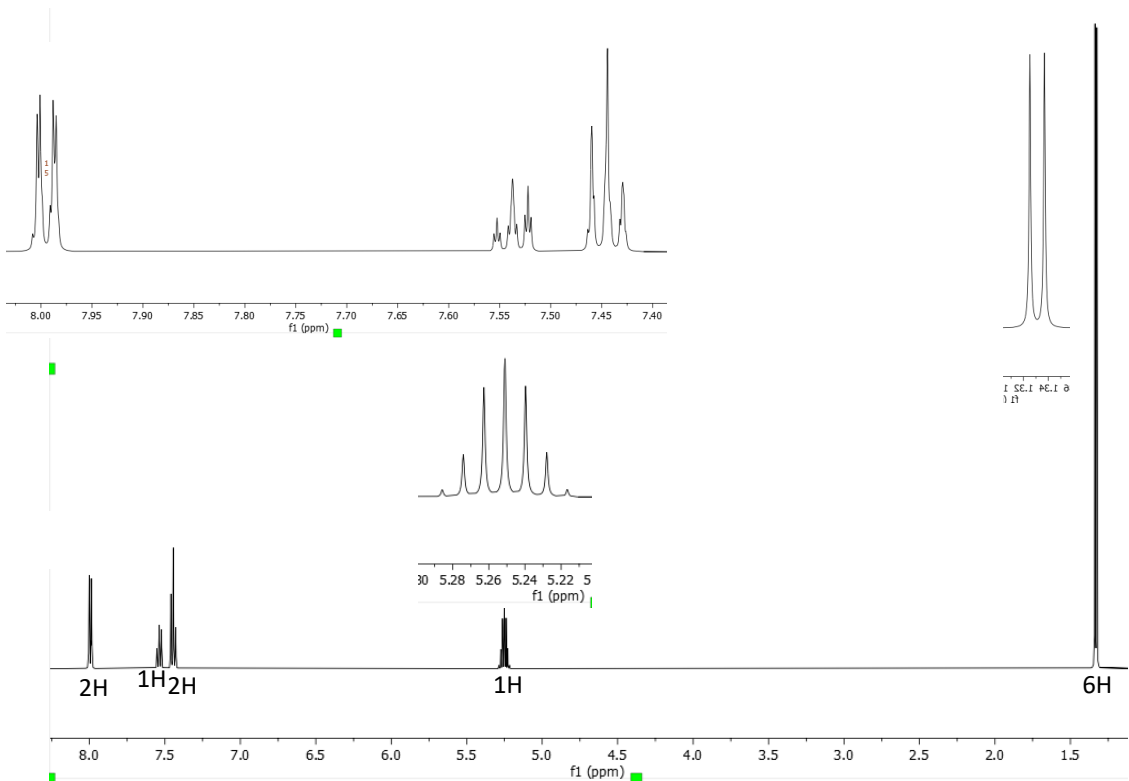
(CSIR NET)

5. ^1H and ^{13}C NMR is given of a compound with molecular formula $\text{C}_{10}\text{H}_{12}\text{O}_2$ and it has one exchangeable proton near 12 ppm . Justify the DEPT results. What is the structure of the compound.



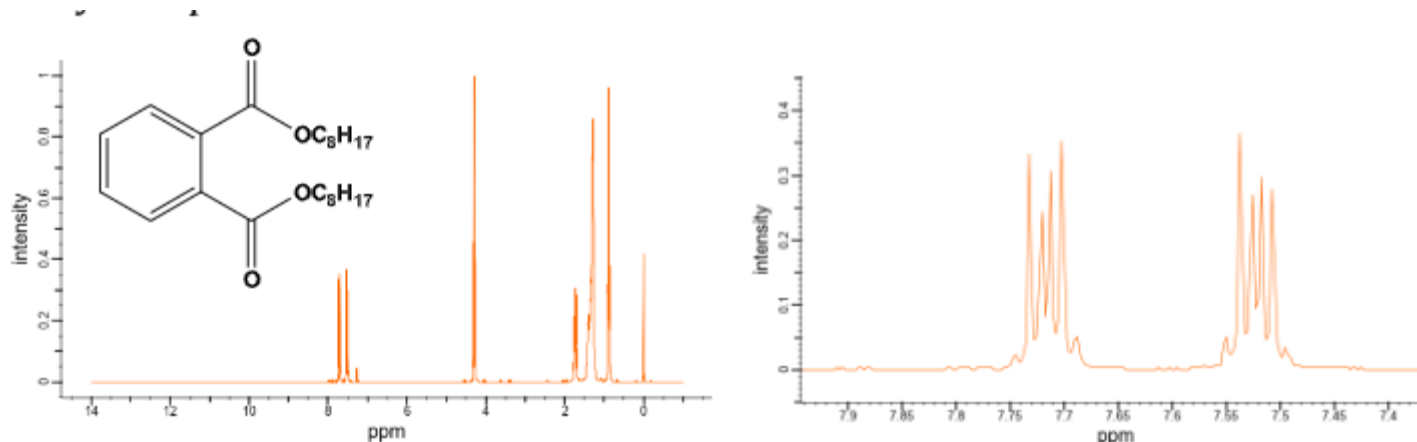
Normal Carbon	DEPT-135	DEPT-90
26	Positive	No peak
34	Positive	Positive
126.6	Positive	Positive
130.5	Positive	Positive

6. Continuing with same molecular formula ($C_{10}H_{12}O_2$) another compound's NMR is given. Suggest the structure and justify the DEPT results.



Normal carbon	DEPT-135	DEPT-90
22	Positive	No Peak
68.36	Positive	Positive
128.5	Positive	Positive
129.7	Positive	Positive
133	Positive	Positive

1. Di-n-octylphthalate is a colorless, odorless, oily liquid that is used to keep plastics soft or more flexible (plasticizer). If you record the ^1H NMR spectra of this compound, you will find that the pattern for the aromatic protons look more complicated than expected. Could you explain the reason behind it?



2. In the ^1H NMR spectrum recorded at 293 K, an organic compound ($\text{C}_3\text{H}_7\text{NO}$) exhibited signals at δ 7.8 (1H, s), 2.8 (3H, s), and 2.6 (3H, s). What is the compound? (CSIR NET)

3. An aromatic compound which is mono-substituted with molecular formula **C_9H_{12}** shows three ^1H NMR signals at δ 7.2 (5H), 2.9 (1H), and 1.2 (6H). Predict its structure and assign the peaks.

4. Predict the splitting pattern of ^{13}C NMR spectrum of $\text{TiCl}_3(\text{CDH}_2)$? Assume that it only couples with H and D, and $^1J(\text{C-H}) > ^1J(\text{C-D})$.

(CSIR NET)

1. An organic compound with molecular formula **C₉H₁₀O₃** exhibited the following spectral data:

IR: 3400, 1680 cm⁻¹

¹H NMR: δ 7.8 (1H, d, J = 8 Hz), 7.0 (1H, d, J = 8 Hz), 6.5 (1H, s), 5.8 (1H, s, D₂O exchangeable), 3.9 (3H, s), 2.3 (3H, s).

Identify the compound.

(CSIR NET)

2. An organic compound having molecular formula C₁₀H₁₂O₂ exhibits the following spectral data:

IR: 3400 (br), 1600 cm⁻¹

¹H NMR (δ, ppm): 1.85 (3H, d, J = 6 Hz), 3.8 (3H, s), 5.0 (1H, s, D₂O exchangeable), 6.0 (1H, dq, J = 18, 6 Hz),

6.28 (1H, d, J = 18 Hz), 6.75 (1H, d, J = 8 Hz), 6.8 (1H, s), 6.90 (1H, d, J = 8 Hz)

¹³C NMR (δ, ppm): 146.5, 144.0, 131.0, 130.5, 123.0, 119.0, 114.0, 108.0, 55.0, 18.0

Determine the structure of the compound.

3. Why we observe peak of DMSO-d₆ in ¹H NMR although there is no 'H' present and why the peak is observed as quintet ?

In ¹³C NMR spectrum it gives a signal at δ 39.7 ppm, what is the multiplicity of the signal?

4. An organic compound **A** (C₈H₁₆O₂) on treatment with an excess of methyl-magnesium chloride generated two alcohols **B** and **C**, whereas reaction of **A** with lithium aluminium hydride generated only a single alcohol **C**. Compound **B** on treatment with an acid yielded an olefin **D** (C₆H₁₂), which exhibited only a singlet at δ =1.5 ppm in the ¹H NMR spectrum. Predict the structure of compound A, B, C and D.

(CSIR NET)

5. Draw the structure of the molecule which molecular formula is $C_5H_{10}O_2$.

1H : 1.10 (s, 6H), 3.5 (s, 4H)

6. A compound with molecular formula $C_4H_{10}O$ has four isomers.

Addition of Anhy. $ZnCl_2$ and Conc. HCl to isomer 1 it gives instant turbid solution whereas for isomer 2 turbidity appears slowly and isomer 3 and 4 shows no turbid solution at RT upon addition of the reagent.

Isomer 1

1H NMR: 1.20 (s, 9H), 4.5 (s, 1H)

^{13}C NMR : 30.9, 70.3

Upon addition of Lucas reagent tertiary alcohol solution becomes turbid instantly, Secondary alcohol reacts slowly and primary alcohol does not react at RT

Isomer 3:

1H NMR: 0.94 (t, 3H), 1.39 (m, 2H), 1.56 (m, 2H), 3.64 (t, 2H), 4.7 (s, 1H)

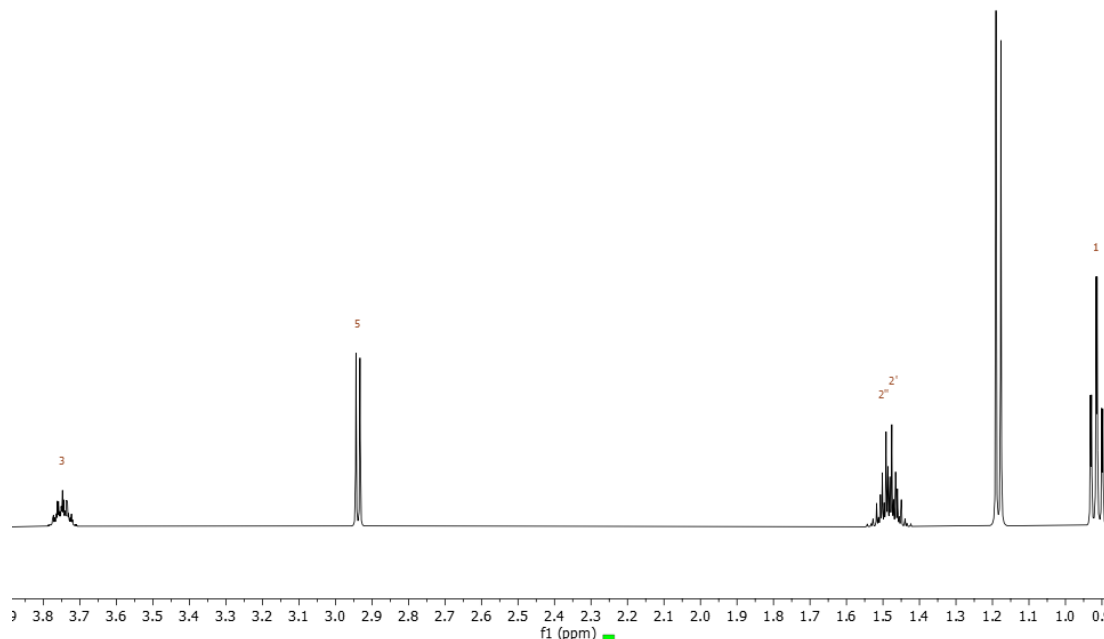
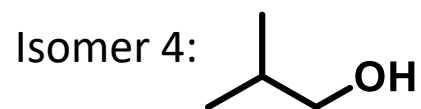
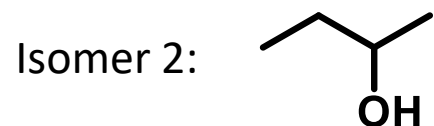
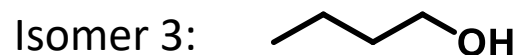
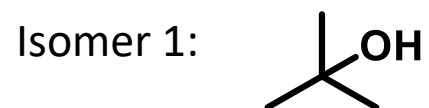
^{13}C NMR: 14.1, 18.7, 34.4, 62.5

Predict the structure of Isomer 1 and 3.

Draw the structures of isomers 2 and 4 with their 1H NMR spectra (approximate δ value in right order of their chemical shift) with labelling the hydrogen and proper splitting.

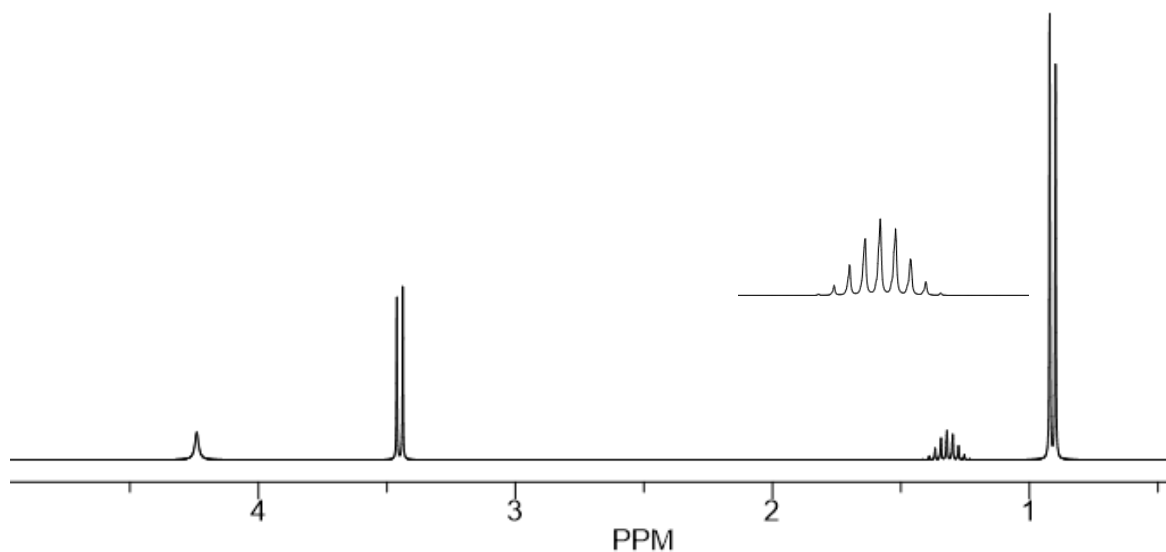
6. Upon addition of Lucas reagent tertiary alcohol solution becomes turbid instantly, Secondary alcohol reacts slowly and primary alcohol does not react at RT

The formula is of butanol



Isomer 2

Diastereotopic H present.



Isomer 4

